

MIT integration bee - 2016

Notación y Consideraciones Generales

A lo largo del documento vamos a utilizar la siguiente notación, en algunos problemas más complejos utilizaremos la notación usual:

$$\{ f(x) \}_a^b := \int_a^b f(x) dx, \quad [f(x)] := \frac{d}{dx} f(x)$$

Recursos obtenidos de: <https://math.mit.edu/~yyao1/integrationbee.html>

Herramientas que te pueden ser útiles:

- Calculadora gráfica: <https://www.desmos.com/calculator?lang=es>
- Calculadora de integrales: <https://mathdf.com/es/>
- Calculadora de integrales: <https://www.wolframalpha.com/>
- Motor de búsqueda para fórmulas en latex <https://approach0.xyz/search/>
- Te ayuda con ideas: <https://chat.deepseek.com>

Ejercicio 1

$$\int \tanh x dx$$

— — — Demostración — — —

$$I = \ln |\cosh(x)| + C$$

Nota:

1. Propiedades básicas de las funciones trigonométricas hiperbólicas:

$$\left\{ \begin{array}{ll} \cosh^2(x) - \sinh^2(x) &= 1 \\ 2 \sinh^2(x) &= \cosh(2x) - 1 \\ 2 \cosh^2(x) &= \cosh(2x) + 1 \\ \sinh^2(x) + \cosh^2(x) &= \cosh(2x) \\ 2 \sinh(x) \cosh(x) &= \sinh(2x) \\ (\sinh(x) + \cosh(x))^2 &= \cosh(2x) + \sinh(2x) \end{array} \right.$$

2. Derivadas de las funciones trigonométricas hiperbólicas:

$$\left\{ \begin{array}{ll} [\sinh(x)] &= \cosh(x) \\ [\cosh(x)] &= \sinh(x) \\ [\tanh(x)] &= \operatorname{sech}^2(x) \\ [\coth(x)] &= -\operatorname{csch}^2(x) \\ [\operatorname{sech}(x)] &= -\operatorname{sech}(x)\tanh(x) \\ [\operatorname{csch}(x)] &= -\operatorname{csch}(x)\coth(x) \end{array} \right.$$

Ejercicio 2

$$\int_{-4}^4 |x^3 - x| dx$$

--- Demostración ---

$$\begin{aligned}
 I &= 2 \left\{ |x| |x^2 - 1| \right\}_0^4 \\
 &= 2 \left(\left\{ |x| |x^2 - 1| \right\}_0^1 + \left\{ |x| |x^2 - 1| \right\}_1^4 \right) \\
 &= 2 \left(- \left\{ x(x^2 - 1) \right\}_0^1 + \left\{ x(x^2 - 1) \right\}_1^4 \right) \\
 &= 2 \left(- \left(\frac{x^4}{4} - \frac{x^2}{2} \right)_0^1 + \left(\frac{x^4}{4} - \frac{x^2}{2} \right)_1^4 \right) \\
 &= 2 \left(- \left(\frac{x^4}{4} - \frac{x^2}{2} \right)_0^1 + \left(\frac{x^4}{4} - \frac{x^2}{2} \right)_1^4 \right) \\
 &= 113
 \end{aligned} \tag{1}$$

- (1): Función par en intervalo simétrico

Ejercicio 3

$$\int_1^e \log(\sqrt{x}) dx$$

--- Demostración ---

$$\begin{aligned}
 I &= \frac{1}{2} \left\{ \ln(x) \right\}_1^e \\
 &= \frac{1}{2} (x \ln(x) - x)_1^e \\
 &= \frac{1}{2}
 \end{aligned}$$

Ejercicio 4

$$\int \left(e^{e^x + e^{-x} + x} - e^{e^x + e^{-x} - x} \right) dx$$

--- Demostración ---

$$\begin{aligned}
 I &= \left\{ e^{e^x + e^{-x}} (e^x - e^{-x}) \right\} \\
 &= \left\{ e^{2 \cosh(x)} (2 \sinh(x)) \right\} \\
 &= e^{2 \cosh(x)} + C
 \end{aligned} \tag{1}$$

- (1): $[e^{f(x)}] = e^{f(x)} f'(x)$

Ejercicio 5

$$\int \frac{\log(\log(x))}{x \log(x)} dx$$

--- Demostración ---

$$I = \left\{ \frac{\log(u)}{u} \right\} \quad (1)$$

$$= \{s\} \quad (2)$$

$$= \frac{s^2}{2} + C$$

$$= \frac{(\ln(\ln(x)))^2}{2} + C$$

$$\bullet (1): \begin{cases} u = \ln(x) \\ du = \frac{1}{x} dx \end{cases}$$

$$\bullet (2): \begin{cases} s = \ln(u) \\ ds = \frac{1}{u} du \end{cases}$$

Ejercicio 6

$$\int_0^{\pi/3} \frac{dx}{1 + \tan^2(x)}$$

— — — Demostración — — —

$$I = \left\{ \frac{1}{\sec^2(x)} \right\}_0^{\pi/3} \quad (1)$$

$$= \{ \cos^2(x) \}_0^{\pi/3}$$

$$= \frac{1}{2} \{ \cos(2x) + 1 \}_0^{\pi/3} \quad (2)$$

$$= \frac{1}{2} \left(\frac{\sin(2x)}{2} + x \right)_0^{\pi/3}$$

$$= \frac{\sqrt{3}}{8} + \frac{\pi}{6}$$

$$\bullet (1): \tan^2(x) + 1 = \sec^2(x)$$

$$\bullet (2): \cos(2x) = 2 \cos^2(x) - 1$$

Ejercicio 7

$$\int_{-27}^{27} \arcsin\left(\frac{x^{1/3}}{3}\right) dx$$

— — — Demostración — — —

$$I = 0 \quad (1)$$

$$\bullet (1): \text{Función impar en intervalo simétrico}$$

Ejercicio 8

$$\int_{50}^{100} \lfloor \log_2 x \rfloor dx$$

— — — Demostración — — —

$$I = \{ \lfloor \log_2 x \rfloor \}_{50}^{64} + \{ \lfloor \log_2 x \rfloor \}_{64}^{100}$$

$$= \{ 5 \}_{50}^{64} + \{ 6 \}_{64}^{100}$$

$$= 286$$

Ejercicio 9

$$\int (e^x \cos x - e^x \sin x) dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \{ e^x (\cos x - \sin x) \} \\ &= e^x \cos(x) + C \end{aligned} \quad (1)$$

- (1): $[e^x f(x)] = e^x (f(x) + f'(x))$

Ejercicio 10

$$\int_0^\infty x^3 e^{-x^2} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ x x^2 e^{-x^2} \right\}_0^\infty \\ &= \frac{1}{2} \{ u e^u \}_0^{-\infty} \\ &= \frac{1}{2} (u e^u - e^u)_0^{-\infty} \quad (2) \\ &= \frac{1}{2} \end{aligned}$$

- (1): $\begin{cases} u &= -x^2 \\ du &= -2x dx \end{cases}$

$$\bullet (2): \left| \begin{array}{c|c} D & I \\ u & e^u \\ 1 & e^u \\ 0 & e^u \end{array} \right|$$

Ejercicio 11

$$\int ((2e^{x^2} x + 1) \cos x - (e^{x^2} + x) \sin x) dx$$

— — — Demostración — — —

$$I = (e^{x^2} + x) \cos(x) + C \quad (1)$$

- (1): $[(e^{x^2} + x)f(x)] = (2e^{x^2} x + 1)f(x) + (e^{x^2} + x)f'(x)$

Ejercicio 12

$$\int (1 + x^{1/2} + x^{1/3})(1 + x^{-1/2} + x^{-1/3}) dx$$

— — — Demostración — — —

$$\begin{aligned}
I &= 6 \left\{ (1+t^3+t^2)(t^5+t^2+t^3) \right\} \\
&= 6 \left\{ t^8+t^7+t^6+3t^5+t^4+t^3+t^2 \right\} \\
&= 6 \left(\frac{t^9}{9} + \frac{t^8}{8} + \frac{t^7}{7} + 3\frac{t^6}{6} + \frac{t^5}{5} + \frac{t^4}{4} + \frac{t^3}{3} \right) + C
\end{aligned} \tag{1}$$

$$\bullet (1): \begin{cases} x &= t^6 \\ dx &= 6t^5 dt \end{cases}$$

Ejercicio 13

$$\int \sin(\sin(x)) \cos(\sin(x)) \cos(x) dx$$

--- Demostración ---

$$I = \left\{ \sin(u) \cos(u) \right\} \tag{1}$$

$$= \frac{1}{2} \left\{ \sin(2u) \right\} \tag{2}$$

$$= -\frac{\cos(2u)}{4} + C$$

$$= -\frac{\cos(2 \sin(x))}{4} + C$$

$$\bullet (1): \begin{cases} u &= \sin(x) \\ du &= \cos(x) dx \end{cases}$$

$$\bullet (2): \sin(2x) = 2 \sin(x) \cos(x)$$

Ejercicio 14

$$\int \left(\frac{\cos(x) + \sin(x)}{x^2} + \frac{\sin(x) - \cos(x)}{x} \right) dx$$

--- Demostración ---

$$\begin{aligned}
I &= \left\{ \frac{\cos(x) + \sin(x)}{x^2} - \frac{\cos(x) - \sin(x)}{x} \right\} \\
&= -\frac{\cos(x) + \sin(x)}{x} + C
\end{aligned} \tag{1), (2)}$$

$$\bullet (1): [\sin(x) + \cos(x)] = \cos(x) - \sin(x)$$

$$\bullet (2): \left[\frac{f(x)}{x} \right] = \frac{-f(x)}{x^2} + \frac{f'(x)}{x}$$

Ejercicio 15

$$\int x^3 \sqrt{x^2 + 1} dx$$

--- Demostración ---

$$\begin{aligned}
I &= \frac{1}{2} \left\{ (u-1)u^{1/2} \right\} \\
&= \frac{1}{2} \left(\frac{2}{5}u^{5/2} - u^{3/2} \frac{2}{3} \right) + C
\end{aligned} \tag{1}$$

$$\bullet (1): \begin{cases} u &= x^2 + 1 \\ du &= 2x dx \end{cases}$$

Ejercicio 16

$$\int \frac{x}{x^4 + x^2 + 1} dx$$

--- Demostración ---

$$\begin{aligned} I &= \frac{1}{2} \left\{ \frac{1}{u^2 + u + 1} \right\} \\ &= \frac{1}{2} \left\{ \frac{1}{(u + \frac{1}{2})^2 + \frac{3}{4}} \right\} \\ &= \frac{2}{3} \left\{ \frac{1}{((u + \frac{1}{2}) \frac{2}{\sqrt{3}})^2 + 1} \right\} \\ &= \frac{1}{\sqrt{3}} \arctan \left(\left(u + \frac{1}{2} \right) \frac{2}{\sqrt{3}} \right) + C \end{aligned} \quad (1)$$

$$\bullet (1): \begin{cases} u &= x^2 \\ du &= 2x dx \end{cases}$$

Ejercicio 17

$$\int e^{e^{2016x} + 6048x} dx$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ e^{e^{\alpha x} + 3\alpha x} \right\} \\ &= \left\{ e^{e^{\alpha x}} e^{\alpha x} e^{2\alpha x} \right\} \\ &= \frac{1}{\alpha} \left\{ e^u u^2 \right\} \quad (1) \\ &= \frac{1}{\alpha} (u^2 e^u - 2u e^u + 2e^u) + C \quad (2) \end{aligned}$$

$$\bullet (1): \begin{cases} u &= e^{\alpha x} \\ du &= \alpha e^{\alpha x} dx \end{cases}$$

$$\bullet (2): \begin{vmatrix} D & I \\ u^2 & e^u \\ 2u & e^u \\ 2 & e^u \\ 0 & e^u \end{vmatrix}$$

Ejercicio 18

$$\int \frac{dx}{1 - x + x^2 - x^3}$$

--- Demostración ---

$$\begin{aligned}
I &= - \left\{ \frac{x+1}{x^4-1} \right\} \\
&= - \left\{ \frac{x+1}{(x-1)(x+1)(x^2+1)} \right\} \\
&= - \left\{ \frac{1}{(x-1)(x^2+1)} \right\} \\
&= - \frac{1}{2} \left\{ \frac{1}{x-1} - \frac{x+1}{x^2+1} \right\} \\
&= - \frac{1}{2} \left\{ \frac{1}{x-1} - \frac{x}{x^2+1} - \frac{1}{x^2+1} \right\} \\
&= - \frac{1}{2} \left(\ln|x-1| - \frac{\ln|x^2+1|}{2} - \arctan(x) \right) + C
\end{aligned} \tag{1}$$

- (1): $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$

Ejercicio 19

$$\int_{\pi/3}^{\pi/2} \frac{1 - \cos x}{\sin x} dx$$

— — — Demostración — — —

$$\begin{aligned}
I &= \left\{ \csc(x) - \cot(x) \right\}_{\pi/3}^{\pi/2} \\
&= \left(-\ln|\csc(x) + \cot(x)| - \ln|\sin(x)| \right)_{\pi/3}^{\pi/2} \\
&= \ln(\sqrt{3}) + \ln(\sqrt{3}/2)
\end{aligned}$$

Ejercicio 20

$$\int_0^\infty \frac{dx}{2 + \cosh x}$$

— — — Demostración — — —

$$\begin{aligned}
I &= \left\{ \frac{1}{1 + 2 \cosh^2(x/2)} \right\}_0^\infty \tag{1} \\
&= \left\{ \frac{1}{\cosh^2(x/2) - \sinh^2(x/2) + 2 \cosh^2(x/2)} \right\}_0^\infty \tag{2} \\
&= \left\{ \frac{1}{3 \cosh^2(x/2) - \sinh^2(x/2)} \right\}_0^\infty \\
&= \left\{ \frac{\operatorname{sech}^2(x/2)}{3 - \tanh^2(x/2)} \right\}_0^\infty \\
&= 2 \left\{ \frac{1}{3 - u^2} \right\}_0^1 \tag{3} \\
&= 2 \left\{ \frac{1}{(\sqrt{3}-u)(\sqrt{3}+u)} \right\}_0^1 \\
&= \frac{1}{\sqrt{3}} \left\{ \frac{1}{\sqrt{3}-u} + \frac{1}{\sqrt{3}+u} \right\}_0^1 \\
&= \frac{1}{\sqrt{3}} \left(-\ln|\sqrt{3}-u| + \ln|\sqrt{3}+u| \right)_0^1 \\
&= \frac{1}{\sqrt{3}} \ln \left| \frac{\sqrt{3}+1}{\sqrt{3}-1} \right|
\end{aligned}$$

- (1): $2 \cosh^2(x) = 1 + \cosh(2x)$

- (2): $\cosh^2(x) - \sinh^2(x) = 1$

$$\bullet (3): \begin{cases} u &= \tanh(x/2) \\ du &= \frac{\operatorname{sech}^2(x/2)}{2} dx \end{cases}$$

Nota:

1. Propiedades básicas de las funciones trigonométricas hiperbólicas:

$$\begin{cases} \cosh^2(x) - \sinh^2(x) &= 1 \\ 2 \sinh^2(x) &= \cosh(2x) - 1 \\ 2 \cosh^2(x) &= \cosh(2x) + 1 \\ \sinh^2(x) + \cosh^2(x) &= \cosh(2x) \\ 2 \sinh(x) \cosh(x) &= \sinh(2x) \\ (\sinh(x) + \cosh(x))^2 &= \cosh(2x) + \sinh(2x) \end{cases}$$

2. Derivadas de las funciones trigonométricas hiperbólicas:

$$\begin{cases} [\sinh(x)] &= \cosh(x) \\ [\cosh(x)] &= \sinh(x) \\ [\tanh(x)] &= \operatorname{sech}^2(x) \\ [\coth(x)] &= -\operatorname{csch}^2(x) \\ [\operatorname{sech}(x)] &= -\operatorname{sech}(x)\tanh(x) \\ [\operatorname{csch}(x)] &= -\operatorname{csch}(x)\coth(x) \end{cases}$$